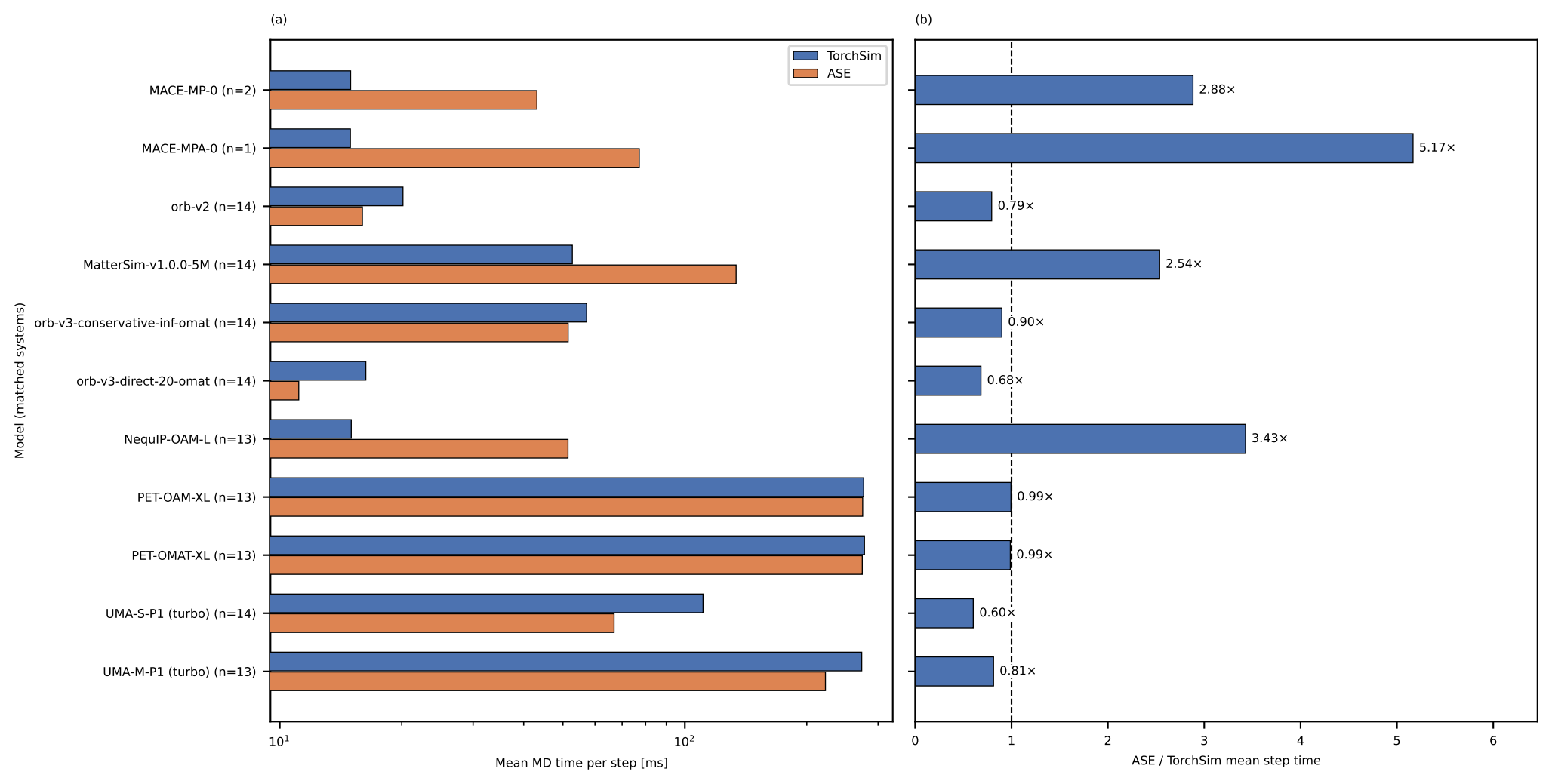
(a)
(b)
TorchSim
ASE
MACE-MP-0 (n=2)
MACE-MPA-0 (n=1)
orb-v2 (n=14)
MatterSim-v1.0.0-5M (n=14)
orb-v3-conservative-inf-omat (n=14)
orb-v3-direct-20-omat (n=14)
NequIP-OAM-L (n=13)
PET-OAM-XL (n=13)
PET-OMAT-XL (n=13)
UMA-S-P1 (turbo) (n=14)
UMA-M-P1 (turbo) (n=13)
Model (matched systems)
10^1
10^2
Mean MD time per step [ms]
2.88×
5.17×
0.79×
2.54×
0.90×
0.68×
3.43×
0.99×
0.99×
0.60×
0.81×
0
1
2
3
4
5
6
ASE / TorchSim mean step time